Huawei Access Node BTM-ACC-34 Hardware/Environmental Fault Root Cause Identification and Resolution

Node: BTM-ACC-34 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_8994

Description

The BTM-ACC-34 node is a critical component of the BTM Layout Aggregation site, serving as a gateway between the external networks and the internal core network. This node is equipped with Huawei's high-performance Access equipment, specifically designed to handle large volumes of traffic with minimal latency and maximum reliability. However, due to the complex nature of modern network infrastructure, hardware failures and environmental faults can still occur, impacting the node's overall performance and leading to the ALARM_8994 (Warning) alarm code being triggered. In this SOP, we will delve into the underlying protocol and hardware mechanics responsible for this failure domain, and outline the necessary diagnostic and resolution steps to rectify the issue.

Pre-requisite Checks

- check: Verify the site location and node ID to ensure BTM-ACC-34 is the correct device, status: Dispatch remote hands to BTM Layout Aggregation to perform the necessary checks

- check: Conduct a redundant link status check to ensure all links are functioning within normal parameters, status: Verify the node's power supply redundancy status

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display alarm-information error, Expected Output: Error information of ALARM_8994 will be displayed, Description: This command will provide detailed information about the ALARM_8994 error, including the timestamp, priority, and cause of the alarm. This information will help in identifying the root cause of the issue.

Command: display hardware info asic, Expected Output: ASIC memory information will be displayed, Description: This command will provide information about the ASIC memory, including the memory size, usage, and any errors that may have occurred. This information will help in identifying if the ASIC memory is the root cause of the issue.
```

```
Command: display power supply status, Expected Output: Power supply status will be displayed, Description: This command will provide information about the power supply status, including the power supply redundancy status and any alarms related to the power supply. This information will help in identifying if the power supply is the root cause of the issue.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: replace asic slot 1/0/1, Description: Replace the faulty ASIC with a new one to rectify the issue. The 'replace asic' command will replace the faulty ASIC with a new one, and the 'slot 1/0/1' specifies the location of the ASIC to be replaced.

Step 2: Step: Replace the faulty line card, Command: replace linecard slot 1/0/0, Description: Replace the faulty line card with a new one to rectify the issue. The 'replace linecard' command will replace the faulty line card with a new one, and the 'slot 1/0/0' specifies the location of the line card to be replaced.

Step 3: Step: Add power supply redundancy, Command: add power supply redundancy, Description: Add power supply redundancy to the node to rectify the issue. The 'add power supply redundancy' command will add power supply redundancy to the node, ensuring that the node remains operational even in the event of a power supply failure.

Post-Check Verification

```
Command: display alarm-information error, Expected Output: ALARM_8994 should not be displayed, Description: This command will verify if the ALARM_8994 is still displayed after the resolution steps. If the ALARM_8994 is still displayed, it indicates that the issue has not been fully resolved.

Command: display hardware info asic, Expected Output: ASIC memory information should be within normal parameters, Description: This command will verify if the ASIC memory information is within normal parameters after the resolution steps. If the ASIC memory information is not within normal parameters, it indicates that the ASIC memory is still faulty.

Command: display power supply status, Expected Output: Power supply status should be within normal parameters, Description: This command will verify if the power supply status is within normal parameters after the resolution steps. If the power supply status is not within normal parameters, it indicates that the power supply is still faulty.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: restore configuration, Description: Restore the original configuration to the node to rectify the issue. The 'restore configuration' command will restore the original configuration to the node, and the node will return to its original state.

- Step: Replace the original faulty component, Command: replace asic slot 1/0/1, Description: Replace the original faulty component with the faulty component that was replaced during the resolution steps. The 'replace asic' command will replace the original faulty component with the faulty component that was replaced during the resolution steps.